PROTOCOL FOR SUBCULTURE OF DIFFERENTIATED BLOOD-BRAIN BARRIER ENDOTHELIAL CELLS ONTO PLATES AND FILTERS

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MATERIALS:

I. Plasticware

FISHER

- Corning Tissue Culture Plates (6- or 12-well, 3513 or 3516)
- 500 mL filter-top bottles (S2GPT05RE)
- Corning Transwell Polyester Filters (07-200-161)

II. Reagents

Thermo Fisher Scientific		
Human endothelial serum-free medium (hESFM)	11111	500 mL
Accutase	A1110501	100 mL
PBS	14190-144	500 mL
B-27 Supplement (50X), serum free	17504-044	10 mL
Peprotech		
Human basic fibroblast growth factor	100-18B	100 ug/mL
Sigma		
All-trans retinoic acid	R2625	50 mg
Fibronectin	F1141	5 mg
Collagen IV	C5533	5 mg

III. Equipment

EVOM2 located in tissue culture room
Chopstick electrodes or EndOhm chamber located in tissue culture room

REAGENT/MEDIUM PREPARATION:

B27 Supplement
Thaw 10 mL bottle and mix thoroughly. Aliquot into sterile microcentrifuge tubes at 280 µL/tube and store at -20ºC. Upon thawing, unused portions of an aliquot may be stored at 4ºC for up to 1 week for further media preparation.

bFGF, 100 µg/mL (prepared according to E8 media protocol)
Thaw a 500 uL aliquot of bFGF and dilute 1:5000 in EC medium for a final concentration of 20 ng/mL as described below. Divide the remaining bFGF in 100 uL aliquots and re-freeze at -80 ºC. These remaining aliquots can be thawed and used for EC medium but cannot be refrozen a second time.

Retinoic acid (RA)
Dilute 50 mg RA in 16.6 DMSO mL to create a stock solution of 10 mM and store 1 mL aliquots at -80 ºC. To prepare working stocks, divide a 1 mL stock tube into 50 uL aliquots and store at -20 ºC. Dilute working stocks 1:1000 in EC medium for a final concentration of 10 µM.

EC medium w/ 200X B27 + 20 ng/mL bFGF
For 50 mL: add 250 µL of B27 and 10 µL bFGF to 50 mL of hESFM.
Good for up to two weeks at 4ºC.

EC medium w/ 200X B27
For 50 mL: add 250 µL of B27 to 50 mL of hESFM.

Note that after subculture, purified BMECs are changed to EC medium containing 200X B27, but without bFGF.

Collagen IV, 1 mg/mL
Dissolve 5 mg of collagen IV in 5 mL of sterile-filtered 0.5 mg/mL acetic acid

ECM solution for plate and filters filters
Mix 5 parts sterile ddH_2O with 4 parts 1 mg/mL collagen IV and 1 part fibronectin. Exact volume depends on the number of filters being coated (200 uL/filter) and number of plates being coated.

SUBCULTURING ONTO PLATES USING ACCUTASE:
Day 6

1. Coat plates with ECM plate solution for at least 1 hour at 37 °C. Volume depends on plate type (see Table):

Plate type for subculture phase	Volume of ECM solution for coating	Working volume of EC media for cell culture
6-well	800 µL	2 mL
12-well	250 µL	1 mL
24-well	200 µL	500 uL
48-well	100 µL	400 uL
96-well	50 µL	200 uL

   **If desired, plates may be coated overnight. If coating overnight, add necessary volume of ECM and an equal volume of ddH_2O to each well to prevent excessive evaporation. If using glass plates, overnight incubation is needed to achieve adequate protein adsorption.
2. Aspirate plates and allow to dry in sterile hood (place the plate in the back of the hood and leave the lid slightly ajar). Plates only need to dry for 5 min (can be aspirated during accutase incubation). Do not over dry!
3. Retrieve cells from incubator and transfer equal volume of spent media to 15 mL conical corresponding to the number of wells being accutased
   a. For example, if accutasing 4 wells, save 4 mL of spent media and discard the rest.
4. Wash each well once with 2 mL PBS.
5. Add 1 mL accutase (warmed to room temperature) to each well.
6. Incubate at 37ºC, length of time depends on cell treatment:
   a. If cells have not been treated with RA, incubate at 37ºC for 20 min, or until cells are dissociated from plate (whichever comes first).
   b. If cells have been treated with RA, incubate at 37ºC for 20 to 25 min typically suffices, though up to 45 minutes may be necessary for some differentiations.
7. Using p1000, collect cells, and spray gently over surface 2-3x to dislodge any remaining cells. Triturate briefly to break up cell clumps.
8. Add cells to 15 mL conical containing spent media.
9. Spin down cells for 4 min at 1000 rpm.
10. Aspirate media, and resuspend cells in appropriate volume of EC media (with or without 10 µM RA). For 6- and 12-wells, cells are seeded based on a split ratio:
    a. 1 well of a 6-well plate is split to 1 well of a 6-well plate [1:1]
    b. 1 well of a 6-well plate is split to 3 wells of a 12-well plate [1:3]
    c. For smaller plates (24-, 48-, or 96-wells), seed 1 million cells/cm^2.
    d. Multiply split ratio by the working volume found in the table to arrive at total volume of EC media (with or without 10 µM RA) in which to resuspend cells.
11. Thoroughly triturate 3-4 times to yield single cell suspension.
12. Add appropriate volume of cells to each well.
13. Place plate in incubator, shaking plate back and forth to distribute cells evenly (do not swirl)
14. 24 hours later (i.e., day 7), aspirate spent media and add appropriate volume of EC medium (**without bFGF or RA**).

SUBCULTURING ONTO <u>FILTERS</u> USING ACCUTASE:
Day 6

Note: owing to the complexity of filter washing and separate addition of media to the top and bottom chambers of filters, make sure you calculate the necessary volumes of EC media before starting the passaging step.

1. Coat Transwell filters with 200 µL of ECM filter solution. Incubate at least 4 h in 37 °C incubator, up to overnight.
2. Aspirate filters and allow to dry in sterile hood, similar to the description for plates. Filters should dry for at least 20 min but no longer than 30 minutes. If cells are still dissociating after 30 minutes, move to step 6a before performing steps 3-5.
3. Retrieve cells from incubator and transfer equal volume of spent media to 15 mL conical corresponding to the number of wells being accutased
    a. For example, if accutasing 4 wells, save 4 mL of spent media and discard the rest.
4. Wash each well once with 2 mL PBS.
5. Add 1 mL accutase (warmed to room temperature) to each well and incubate at 37^{o}C. The length of time depends on cell treatment:
    a. <u>If cells have not been treated with RA</u>, incubate at 37 ^{o}C for 20 min, or until cells are dissociated from plate (whichever comes first). Do not go over 20 minutes.
    b. <u>If cells have been treated with RA</u>, incubate at 37 ^{o}C for 20-45 min, or until cells are dissociated from plate.
6. During accutase incubation, prepare filters:
    a. Add 0.5 mL hESFM to the top of each filter for approximately 5 minutes to wet filters.
    b. After 5 minutes, aspirate hESFM from the top of each filter and add 1.5 mL EC medium (with or without 10 µM RA) to the basolateral chamber of each filter.
7. Following accutase incubation, use p1000 to collect cells and spray gently over surface 2-3x to dislodge any remaining cells. Triturate briefly to break up cell clumps.
8. Add accutased cells to the conical containing spent medium. Spin down for 4 min at 1000 RPM.
9. Aspirate media, resuspend cells in appropriate volume of EC media (with or without 10 µM RA). Cells are seeded based on a 1:3 ratio (i.e. 1 well of a six well plate seeds three Transwell filters).
    a. Volume of EC media containing cells = # of wells of a 6 well plate being subcultured * 0.5 mL
    b. For 3 filters, resuspend cells in 1.5 mL EC medium, for 6 use 3 mL, etc.
10. Place plate in incubator, shaking plate back and forth to distribute cells evenly (do not swirl)
11. 24 hours later (i.e., day 7), aspirate and change to EC medium **lacking bFGF or RA**.
    a. Aspirate EC media from both chambers (do not touch monolayer with pipet tip!).
    b. Gently add EC medium **lacking bFGF or RA**, 0.5 mL in top chamber and 1.5 mL in bottom chamber. **DO NOT include bFGF or RA in the medium** – these factors prevent the monolayer from tightening.

Additional notes:

- Cells should reach confluence 24-48 hours post-split.
- If not treated with RA, cells should reach baseline TEER reading of 200-250 ohms (empty filter not subtracted) when confluent. If retinoic acid is used, initial TEER at 24 hours post-split can range anywhere from 200-2000 ohms depending on the cell fidelity. If using RA, TEER can reach 2000-4000 ohms, depending on the cell line used.

- If using filters for permeability screens, 48 hours post-split is usually the optimal time (maximum TEER).

TROUBLESHOOTING (subculture onto plates and filters)

Problem: Large clumps of cells are accumulating on top of monolayer
Solutions:
1. Increase trituration, particularly if you are not triturating vigorously. Cell clumps can accumulate on the top of the monolayer and decrease viability, while dispersed cells are more likely to remain in solution and can be removed the following day during the medium change.
2. Try increasing length of enzyme treatment (i.e., if using 15 minutes accutase try 20 minutes). Singularized cells (as opposed to clumps of cells) are more likely to form an even monolayer.
3. Try adjusting initial cell density at start of differentiation (day 0). Clumps can indicate that initial cell density was too high.

Problem: Cells have not filled in (resulting in low- to zero TEER if using filters)
Solutions:
1. Treat cells more gently, too much trituration can hurt the cells (particularly if not using RA treatment, these cells tend to be more fragile).
2. Decrease enzyme treatment length (i.e., if using 15 minutes, try 10 minutes). If using accutase, try versene instead.
3. Try adding more cells to the filter. If using accutase, increase number of cells added to filter (i.e., if using 1 million cells try using 1.2 million cells).
4. Try adjusting initial cell density at start of differentiation (day 0). Low attachment to plates/filters can indicate that initial cell density was too low.

Problem: Monolayer is intact but TEER is not spiking, even with RA addition
Solution: Ensure that media is changed after 1 day post-split, and does not contain RA or bFGF. Addition of either RA or bFGF eliminates any potential spike in TEER.